MMV SPICE $\min \| R^{-\frac{1}{2}}(\hat{R}-R)\hat{R}^{-\frac{1}{2}} \|_F^2$

$\hat{R}_{M\times N} = \frac{1}{L} Y_{M\times L} Y^H$

$R_{M\times N} = APA^H + \sigma^2 I = \tilde{A}\tilde{P}\tilde{A}^H$

$\tilde{A}_{M\times(N+M)} = [A \quad I_M] \qquad \tilde{P}_{N+M} = \mathrm{diag}(P_N, \sigma^2 I_M)$

$f = \| R^{-\frac{1}{2}}(\hat{R}-R)\hat{R}^{-\frac{1}{2}} \|_F^2$

$= \mathrm{tr}[R^{-1}(\hat{R}-R)\hat{R}^{-1}(\hat{R}-R)]$

$= \mathrm{tr}(R^{-1}\hat{R}) + \mathrm{tr}(\hat{R}^{-1}R) - 2M$

$= \mathrm{tr}(\hat{R}^{\frac{1}{2}}R^{-1}\hat{R}^{\frac{1}{2}}) + \sum_{n=1}^{N} \tilde{p}_n \tilde{a}_n^H \hat{R}^{-1}\tilde{a}_n - 2M$

为什么 SPICE 是 SDP 问题

$\min_R \langle R, \hat{R}^{-1}\rangle + \langle R^{-1}, \hat{R}\rangle \qquad \frac{x}{a} + \frac{a}{x} \qquad x \geq 0$

s.t. $R \geq 0$

Asymtotical $L \to \infty$, $\hat{R} \to R$, $\mathrm{tr}(\hat{R}^{-1}R) \to M$

所以 $L \to \infty$ 时加上约束

$\min_{\tilde{P}} \mathrm{tr}(\hat{R}^{\frac{1}{2}}R^{-1}\hat{R}^{\frac{1}{2}}) \quad$ s.t. $\sum_{n=1}^{N+M} \tilde{w}_n \tilde{p}_n = 1 \qquad \tilde{w}_n = \frac{1}{M}\tilde{a}_n^H \hat{R}^{-1}\tilde{a}_n$

转化为 SDP 问题

$\min_{\tilde{P}} \mathrm{tr}(C^H \tilde{P}^{-1} C) \quad$ s.t. $\underset{(N+M)\times M}{\tilde{A}C} = \hat{R}^{\frac{1}{2}}$, $\sum_{n=1}^{N+M} \tilde{w}_n \tilde{p}_n = 1$

Pf: $\mathrm{tr}(\hat{R}^{\frac{1}{2}}R^{-1}\hat{R}^{\frac{1}{2}})$

$\mathrm{tr}(\hat{R}^{\frac{1}{2}}R^{-1}RR^{-1}\hat{R}^{\frac{1}{2}})$

$\mathrm{tr}(\hat{R}^{\frac{1}{2}}R^{-1}\tilde{A}\tilde{P}\tilde{A}^H R^{-1}\hat{R}^{\frac{1}{2}})$

$\mathrm{tr}(\hat{R}^{\frac{1}{2}}R^{-1}\tilde{A}\tilde{P}\tilde{P}^{-1}\underbrace{\tilde{P}\tilde{A}^H R^{-1}\hat{R}^{\frac{1}{2}}}_{C})$

最优解：$C^* = \tilde{P}\tilde{A}^H R^{-1}\hat{R}^{\frac{1}{2}}$

最优值：$\mathrm{tr}(\hat{R}^{\frac{1}{2}}R^{-1}\hat{R}^{\frac{1}{2}})$

Pf: $C^H\tilde{P}^{-1}C \geq (C^*)^H\tilde{P}^{-1}C^* = \hat{R}^{\frac{1}{2}}R^{-1}\hat{R}^{\frac{1}{2}}$

Since $\begin{bmatrix} C^H\tilde{P}^{-1}C & \hat{R}^{\frac{1}{2}} \\ \hat{R}^{\frac{1}{2}} & R \end{bmatrix} \geq 0$

迭代算法

1) 当不同噪声

$C = \tilde{P}\tilde{A}^H R^{-1}\hat{R}^{\frac{1}{2}}$ $\qquad C = \begin{bmatrix} C_1 \\ \vdots \\ C_{N+M} \end{bmatrix}$

再给定 C 后 $\min\limits_{\tilde{P}} tr(C^H\tilde{P}^{-1}C) = \sum\limits_{n=1}^{N+M} \frac{\|C_n\|^2}{\tilde{P}_n}$ $\quad$ s.t. $\sum\limits_{n=1}^{N+M} \tilde{w}_n\tilde{P}_n = 1$

$\tilde{P}_n = \frac{1}{\rho}\frac{\|C_n\|}{\sqrt{w_n}}$ $\qquad \rho = \sum\limits_{n=1}^{N+M}\sqrt{w_n}\|C_n\|$

2) 当相同噪声

$C = \tilde{P}\tilde{A}^H R^{-1}\hat{R}^{\frac{1}{2}}$

$\min\limits_{\tilde{P}} tr(C^H\tilde{P}^{-1}C) = \sum\limits_{n=1}^{N+M}\frac{\|C_n\|^2}{\tilde{P}_n}$ $\quad$ s.t. $\sum\limits_{n=1}^{N+M}\tilde{w}_n\tilde{P}_n = 1$, $\tilde{P}_{N+1} = \cdots = \tilde{P}_{N+M} = \sigma^2$

$\min\limits_{P,\sigma^2} tr(C^H\tilde{P}^{-1}C) = \sum\limits_{n=1}^{N}\frac{\|C_n\|^2}{P_n} + \sum\limits_{n=N+1}^{N+M}\frac{\|C_n\|^2}{\sigma^2}$ $\quad$ s.t. $\sum\limits_{n=1}^{N}\tilde{w}_n P_n + \underbrace{\sum\limits_{n=N+1}^{N+M}\tilde{w}_n}_{:=\gamma}\sigma^2 = 1$

$P_n = \frac{1}{\rho}\frac{\|C_n\|}{\sqrt{w_n}}$ $\qquad \sigma^2 = \frac{1}{\rho}\frac{1}{\sqrt{\gamma}}\left[\sum\limits_{n=N+1}^{N+M}\|C_n\|^2\right]^{1/2}$

$\rho = \sum\limits_{n=1}^{N}\sqrt{w_n}\|C_n\| + \sqrt{\gamma}\left[\sum\limits_{n=N+1}^{N+M}\|C_n\|^2\right]^{1/2}$

SPARROW

$\min \ \frac{1}{2}\|Y - AX\|_F^2 + \lambda\sqrt{N}\|X\|_{2,1}$

$\min \ tr[(ASA^H + \lambda I)^{-1}\hat{R}] + tr(S)$ $\qquad R = APA^H + \lambda I := \tilde{A}\tilde{P}\tilde{A}^H$

$\min \ tr(\hat{R}^{\frac{1}{2}}R^{-1}\hat{R}^{\frac{1}{2}}) + tr(S)$

$\min \ tr(\hat{R}^{\frac{1}{2}}R^{-1}\hat{R}^{\frac{1}{2}})$ $\quad$ s.t. $\sum P_k = 1$ 不行，没有渐近性质

$\min \ tr(C^H\tilde{P}^{-1}C) + tr(\tilde{P})$ $\quad$ s.t. $\tilde{A}C = \hat{R}^{\frac{1}{2}}$

Pf: $C^H\tilde{P}^{-1}C \geq \hat{R}^{\frac{1}{2}}R^{-1}\hat{R}^{\frac{1}{2}}$, then $tr(C^H\tilde{P}^{-1}C) + tr(\tilde{P}) \geq tr(\hat{R}^{\frac{1}{2}}R^{-1}\hat{R}^{\frac{1}{2}}) + tr(\tilde{P})$

迭代算法

$C = \tilde{P}\tilde{A}^H R^{-1}\hat{R}^{\frac{1}{2}}$

$\min \ \sum\limits_{n=1}^{N}\frac{\|C_n\|^2}{P_n} + \sum\limits_{n=1}^{N}P_n$, $\quad P_n = \|C_n\|$